RISHABH KUMAR

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Senior Data Scientist & AI Leader | ICML 2025 Best Paper Winner

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Senior Data Scientist and Machine Learning Engineer with **7+ years of experience** architecting production AI/ML foundations across identity resolution, behavioral representation learning, and synthetic consumer simulation. Specialized in **foundation representation learning, transformer architectures (cross-attention, mask-aware), LLM calibration, and deep generative models** across **350M+ users**. Published author (**ICML 2025 Best Paper**, NeurIPS 2025 Workshop) on modeling user behavior from adaptive surveys. Proven track record modernizing legacy distributed systems and shipping scalable deep learning pipelines on AWS SageMaker and Apache Spark.

TECHNICAL EXPERTISE & CORE COMPETENCIES

AI & Deep Learning:	Transformers (Cross-Attention, Mask-Aware, Self-Attention), Representation Learning, Self-Supervised Learning, Autoencoders, Deep Matrix Factorization, Multi-Label Classification, Probabilistic Modeling, PU Learning, Anomaly Detection.
Generative AI & Agents:	Synthetic Audience Simulation, Agent Persona Parameterization, LLM Calibration (Isotonic Regression, Conformal Prediction, Distribution Alignment), LLMOps & Evaluation, In-Context Persona Grounding, LangChain / LangGraph, RAG.
Languages & Frameworks:	Python, PyTorch, TensorFlow/Keras, Scala, SQL/PostgreSQL, R, Hugging Face, Spark ML, XGBoost, LightGBM, scikit-learn, Optuna, MLflow, TensorBoard, Docker, Git/GitHub, CI/CD.
Cloud & Distributed Data:	Apache Spark, AWS Ecosystem (SageMaker, EMR, Batch, Step Functions, S3, EC2, Lambda, SQS, RDS), Grafana, PII Masking, Consent-Based Consumer Data Pipelines, Privacy-Preserving Identity Resolution.

PROFESSIONAL EXPERIENCE

Resonate Networks

Herndon, VA (Hybrid / Boston, MA)

Senior Data Scientist

`Apr 2024 – Present`

- **Technical Lead for Keystone:** Served as technical lead for Keystone, coordinating a cross-functional team of 6 across data engineering, product, DS/ML, and executive stakeholders to drive technical direction and execution, moving identity intelligence from transient ID-level behavior toward durable person-level representations.
- **Self-Supervised Behavioral Autoencoder (BAE):** Developed a self-supervised transformer/autoencoder foundation model deployed on AWS SageMaker, compressing geo, domain, and NLP-derived category signals into **512-dimensional behavioral embeddings** across millions of daily events for **350M+ users**.
- **Large-Scale Survey Imputation & Persona Reconstruction:** Built a downstream supervised Deep Matrix Factorization model that consumes BAE embeddings to predict **~19K survey attributes (+0.07 mean AUC lift)**, reconstructing sparse adaptive survey responses with population-level calibration across 570K respondents — powering core audience segmentation products and campaign targeting.
- **ICML 2025 Best Paper Architecture (LANTERN):** Formulated and productionized a modular late-fusion cross-attention neural architecture fusing sparse user survey priors with external behavioral features via learned bimodal gating; achieved **+5 point recall lift** and strong performance on rare consumer preferences, directly preserving population heterogeneity.
- **Media Bubble & Ideological Cohort Clustering:** Built *Media Bubble*, an unsupervised behavioral clustering pipeline classifying **~20M devices** into ideological consumption cohorts using real-time behavioral event signals, directly informing campaign creative messaging and voter landscape strategy.

ML Engineer → ML Engineer II → Senior ML Engineer

`Nov 2019 – Mar 2024`

- **Lookalike Audience Discovery (Positive-Unlabeled Learning):** Originated a new lookalike-audience product by adapting an existing pipeline for a new use case, using positive-unlabeled learning to identify high-propensity prospects across **~500M HEM profiles**; outperformed production in **~90% of evaluated audiences** on recall-lift and cumulative-gain metrics.
- **Distributed Spark/Scala NLP Pipeline Modernization:** Migrated a legacy MapReduce NLP pipeline to a distributed Apache Spark/Scala architecture processing millions of daily internet events; reduced compute costs by **~35%** and accelerated processing runtime by **~2.8x**.
- **Large-Scale Production Orchestration:** Streamlined and productionized automated training, inference, and monitoring workflows across AWS SageMaker, EMR, AWS Batch, S3, Step Functions, and SQS for **300M+ active user profiles** across ~1,800 XGBoost models.

HiLabs Inc.

Germantown, MD

Data Scientist

`Mar 2019 – Oct 2019`

- Developed risk adjustment and anomaly detection products for healthcare data using frequent-pattern mining on **~3.5B records**, translating high-dimensional data into strategic insights and client-facing recommendations.

PERSONAL AI PROJECTS

HouseRank — Multi-Agent AI Real Estate Decision & Ranking Platform

`Next.js • LangGraph Multi-Agent • OpenAI API • Framer Motion`

- Architected a multi-agent evaluation pipeline coordinating specialized agents (Listing Intelligence, Financial/Investment Modeling, and Lifestyle Fit) with a synthesis evaluator to score properties across **Personal Fit** and **Investment Potential** (PITI+H true affordability & equity growth).
- Built an interactive in-person tour module enabling buyers to log and weight qualitative on-site observations (natural lighting, acoustic/noise levels, layout flow) that trigger dynamic agent re-scoring across aggregate rankings.

Siyu Chef — Multimodal AI Baby Food Recommendation & Grocery Fulfillment (Android Beta)

`Android • FastAPI • Multimodal Vision • SQLite • Amazon PA-API`

- Developed an Android app (in beta) generating stage-appropriate, nutritionally balanced infant meal recommendations personalized to developmental age, nutritional requirements, and **strict allergen/dietary profiles**.
- Built a multimodal AI vision scanner that detects on-hand fridge/pantry ingredients to suggest instant recipes, paired with a FastAPI/SQLite backend powering automated **1-click Amazon Cart grocery fulfillment** via PA-API.

PUBLICATIONS

Modeling User Behavior from Adaptive Surveys with Supplemental Context

Best Paper Award

Aman Shukla, Daniel Patrick Scantlebury, Rishabh Kumar — **ICML 2025 New in ML Workshop** (arXiv:2507.20919)

Responsible Imputation of User Behavior Surveys via Mask-Aware Transformers

`NeurIPS 2025`

NeurIPS 2025 Workshop on Reliable ML from Unreliable Data

EDUCATION

Carnegie Mellon University (CMU) — Pittsburgh, PA | 2017 – 2018

BITS Pilani — **B.E. Chemical Engineering & M.Sc. Biological Sciences (Dual Degree)** — Pilani, India | 2011 – 2016

HONORS & AWARDS

Best Paper Award: ICML 2025 New in ML Workshop (July 2025)

Reed Award Winner: Best Use of Data Analytics / ML in Online Fundraising (Feb 2026)

1st Prize Winner: Resonate Networks Annual Hackathon (June 2021 & July 2025)